NIT [CAMPUS] · 42-HOUR HACKATHON

# Planetary Propulsion Simulation Hackathon

Participant guide: how to build a real-life simulation of artificial planetary propulsion, such as the Octo-Earth concept

**APPLICATIONS CLOSE**
24 October 2026

**BUILD WINDOW**
42 hours, 24 to 26 October 2026

**PRIZES**
₹30,000 to ₹1,00,000 and 1% equity, awarded on 26 October

**CAREERS**
Full-time, part-time or internship offers based on individual contribution

**TEAMS**
2 to 4 members

**FORMAT**
Software only

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*Engineering the impossible of tomorrow.*

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## How to use this guide

This guide helps your team plan the 48 hours and get the physics right. The rulebook sets the rules; where the two differ, the rulebook wins.

1. **The challenge in one page**: what you build and what it takes to win
2. **The physics you need**: gravity, orbits, momentum, the Lorentz force, structures, climate
3. **The numbers that decide everything**: energy, force and time, worked out
4. **Recommended architecture**: how to structure a simulation that judges trust
5. **A 48-hour plan**: hour-by-hour milestones
6. **Validation**: running the ten benchmarks
7. **Hypothetical enablers and the energy bonus**
8. **Common mistakes** that cost marks
9. **Presenting and submitting**: checklist and judging summary
10. **Resources**

> **The one idea to remember**
>
> A planet only changes its orbit if something *outside* the Earth system pushes on it. Every force your machine makes must be traced to the body that receives the equal and opposite push. If that body is part of Earth, the force moves nothing.

# 1. The challenge in one page

Your team builds a computer simulation of a machine that tries to move Earth to a new orbit, for example from 1 AU out to about 2.7 AU near the asteroid belt, without harming the planet or life on it.

### The reference concept: Octo-Earth

- A structural ring built around Earth along a great circle.
- Eight articulated arms rooted on the ring at equal 45° spacing, all at the same mounting radius.
- Arms up to about 12,000 km long, with segments, hinges, motors and a gripper at each tip.
- Conductors in the arms carry current that interacts with magnetic fields through the Lorentz force.
- The arms extend together on coordinated trajectories and keep the system balanced.

You may simulate Octo-Earth, change it, or propose another concept that acts on a planet-sized body: a gravity tractor, steered asteroid flybys, a mass driver, a solar sail or mirror array, a magnetic sail.

### To win, your concept must work

| Winning condition (all required) | What it means in practice |
|---|---|
| Earth reaches its target orbit | Within 10,000 simulated years |
| Earth stays safe | Every threshold in the rulebook appendix holds for the whole run |
| The structure survives | Every structural safety factor stays at or above 1.5 |
| The code is validated | Every applicable benchmark passes |
| The physics is honest | Every force booked with its reaction; energy and momentum conserved |
| No unresolved flags | Only flags caused by your declared hypothetical enablers are allowed |

You may declare up to three hypothetical enablers (section 7). Entries that honestly show their concept failing are still scored and can win other awards.

# 2. The physics you need

## 2.1 Gravity and orbits

Each body accelerates toward every other body. Model at least the Sun, Earth and Moon; adding the planets costs little.

$$a_i = \Sigma_{j \neq i}\, G\, m_j\, (r_j - r_i) \,/\, |r_j - r_i|^3$$

An orbit's size is its semi-major axis *a*, fixed by the specific orbital energy ε. Raising *a* means adding energy.

$$\varepsilon = v^2/2 - \mu/r = -\mu/(2a) \quad \cdot \quad da/dt = 2a^2\, (v \cdot a_{push}) \,/\, \mu$$

The second formula (Gauss's equation) tells you how fast a small push along the velocity raises the orbit. Use it to check your integrator.

## 2.2 Momentum and reaction forces

Newton's third law: every force has an equal and opposite partner. The momentum of a closed system never changes. So when the arms push on something, ask what that something is attached to.

| The arm pushes on… | Reaction lands on | Moves Earth? |
|---|---|---|
| Earth's own magnetic field | Currents in Earth's liquid core | No: internal |
| Another arm, or the ring | The same structure | No: internal |
| Ionosphere or magnetospheric plasma | Plasma held by Earth's gravity and field | Unclear: must be modelled |
| Solar-wind or interstellar plasma beyond the magnetosphere | Plasma carried away into space | Yes, if the circuit truly closes there |
| Sunlight, an asteroid's gravity, ejected mass | Photons, the asteroid, the ejecta | Yes |

## 2.3 The Lorentz force

$$F = I\, L \times B \quad (\text{integrate } dF = I\, dl \times B \text{ along each conductor})$$

- Current needs a closed path. A closed loop in a *uniform* field feels **zero** net force; only field gradients or an external return path give a net push.
- Evenly spaced arms cancel each other's push in a uniform field. Symmetry that is good for balance is bad for thrust: work out what that means for your design.
- Fields have sources. A field is not a wall to push against; its momentum partner is the current or plasma that carries it.

### 2.4 Structures

Long arms fail long before they produce useful thrust. Check every segment and hinge.

> $\sigma = N/A \pm M\,c/I$ · safety factor = allowable stress / actual stress · $P_{cr} = \pi^2 EI/l^2$

- Loads: weight (gravity falls with altitude), Earth's rotation (centrifugal and Coriolis effects), inertia of moving joints, Lorentz forces, heat.
- A rough limit: a column of any material can stand only about σ/(ρg) tall under its own weight. For steel that is a few kilometres; even ideal nanotubes stop near a few thousand.
- A 12,000 km arm is longer than Earth's radius: it is no longer a small attachment. Gravity varies along it and its natural vibration period can be hours or years.

### 2.5 Keeping Earth habitable

Sunlight falls with the square of distance, so a farther orbit is colder.

> $S = L_\odot / (4\pi r^2)$ · $T_{eq} = [S\,(1 - A) / (4\sigma)]^{1/4} \approx 254\ \mathrm{K} \times (1\ \mathrm{AU} / r)^{1/2}$

At 2.7 AU the equilibrium temperature falls to about 155 K, so without some plan for heating, the oceans freeze. Your simulation must show, not assume, what happens to temperature, liquid water, air, magnetic shielding and the Moon.

### 2.6 Numerical precision

- Use 64-bit (double) numbers for orbits and fields. Earth's distance from the Sun, $1.5 \times 10^{11}$ m, is resolved only to about $3 \times 10^{-5}$ m in double precision, and to about 10 km in 32-bit floats.
- Tiny pushes vanish when added to huge positions. Track the *change* from an unpushed reference orbit separately (Encke's method) to keep them.
- Use a symplectic integrator (leapfrog, Yoshida) for long orbit runs, or an adaptive one (Dormand–Prince, `solve_ivp`) with tight tolerances. Halve the time step and show the answer does not change.
- Run fast processes (vibration, circuits, heat) on their own shorter time step.

## 3. The numbers that decide everything

Work these out in the first hours. They tell you which designs are worth simulating.

| Quantity | Value | How it is found |
|---|---|---|
| Energy to raise Earth–Moon from 1 AU to 2.7 AU | $\approx 1.7 \times 10^{33}$ J | $\Delta E = \mu m/2\ (1/a_1 - 1/a_2)$ |
| …the same, in days of the Sun's entire output | ≈ 50 days | $\Delta E$ / $3.828 \times 10^{26}$ W |
| Velocity change, slow spiral | ≈ 11.6 km/s | $\lvert v_1 - v_2 \rvert$ for circular orbits |
| Hohmann transfer (two burns) | ≈ 6.2 + 4.8 km/s | Standard transfer formulas |
| Average power to finish within 10,000 years | $\approx 5.4 \times 10^{21}$ W | $\Delta E$ / 10,000 years |
| Constant force needed for that spiral | $\approx 2.2 \times 10^{17}$ N | m $\Delta v$ / time |
| Lorentz force, 1 MA × 12,000 km × 5 nT | ≈ 60 N | F = I L B |

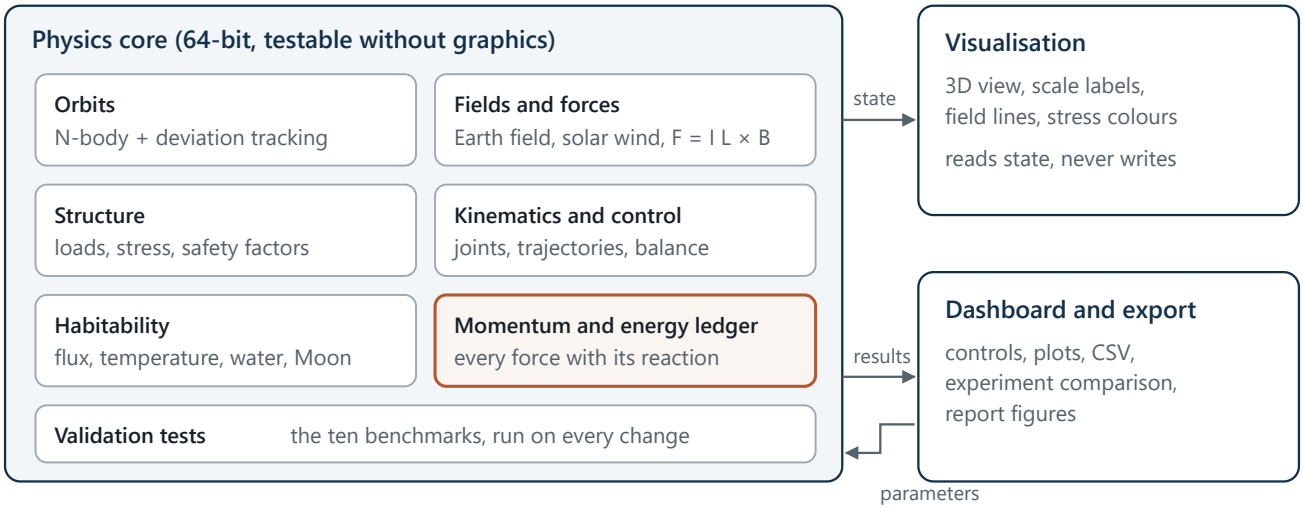
**What the table says**

A concept producing tens of newtons needs about $10^{15}$ times more force. Do this estimate first, decide which hypothetical enablers you need, and design around the gap rather than discovering it on day two.

Values are rounded estimates for planning. Your simulation must compute its own.

# 4. Recommended architecture

Keep the physics separate from the graphics. Judges trust a simulation whose core can run, and be tested, without any visuals.



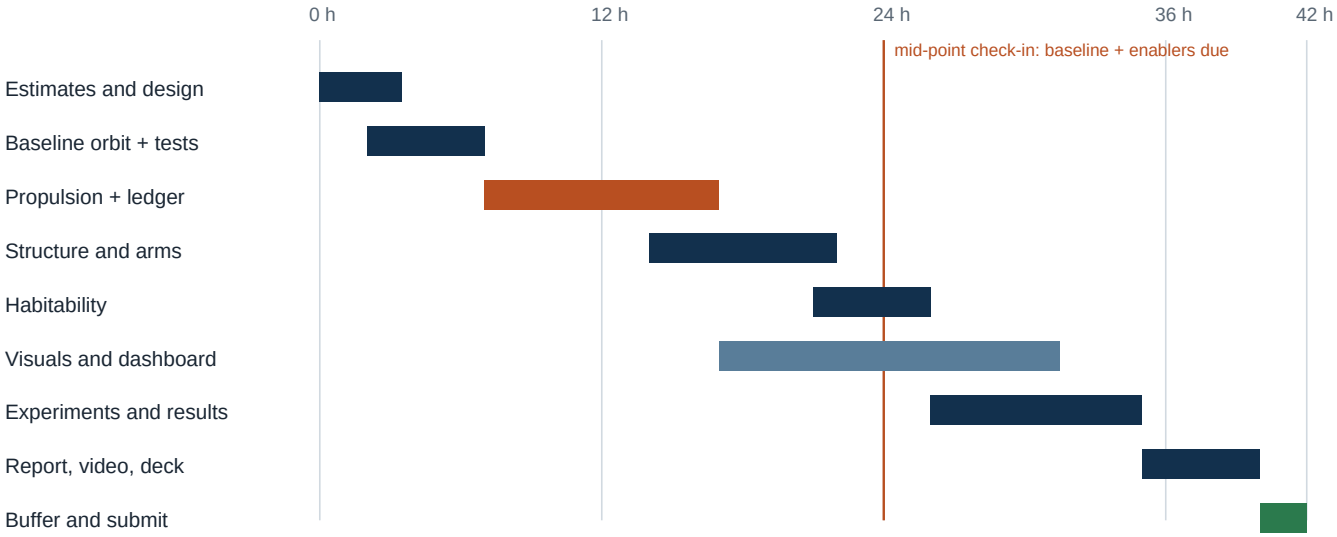

The ledger is the heart of the simulation: it is how you prove which forces can move Earth. Graphics may use a game engine; the core must not depend on it.

## Choosing a stack

| Option | Good for | Watch out for |
|---|---|---|
| Python (NumPy, SciPy, REBOUND) | Fast physics prototyping, plots, validation | Slow inner loops: vectorise or use Numba |
| C++ or Julia | Speed for long runs and many segments | Less time left for visuals |
| JavaScript with three.js | Interactive 3D in the browser; easy to demo | Use plain JS numbers (64-bit) for physics, not GPU floats |
| Unity or Unreal | Polished visuals, joint mechanics up close | Built-in physics is 32-bit and has no electromagnetism |
| Blender | Modelling arms, ring and Earth assets | An asset tool, not a physics engine for this task |

## 5. A 42-hour plan

Have a validated baseline orbit by hour 8 and a working ledger by hour 20. Everything after that is depth.




| Hours | Goal | Done when |
|---|---|---|
| 0–4 | Estimates, concept choice, enablers drafted, roles split | Section 3 numbers computed; one-page design |
| 2–7 | Sun–Earth–Moon baseline with real initial conditions | Benchmarks 4–6 pass; energy drift reported |
| 7–17 | Propulsion model and momentum ledger | Every force has a reaction partner; benchmarks 1–3, 7, 8 pass |
| 14–22 | Arms, joints, loads, safety factors | Benchmark 9 passes; first failing part identified |
| 21–26 | Habitability along the trajectory | Benchmark 10 passes; thresholds checked |
| 17–31 | 3D view, controls, plots | Scale labels on screen; parameters adjustable |
| 26–35 | Experiments: baseline, propulsion, failure envelope | Results saved and compared |
| 35–40 | Report, 3-minute video, 7-minute deck | All six deliverables drafted |
| 40–42 | Buffer, final run, submit | Submitted before the deadline on 26 October |

**Splitting a team of four**

One person owns orbits and validation, one owns forces and the ledger, one owns structure and habitability, one owns visuals, report and pitch. Everyone reviews the ledger.

# 6. Validation

Write each benchmark as an automated test and run all of them after every major change. Judges may change the inputs during judging.

| # | Test | Expected | Tolerance |
|---|---|---|---|
| 1 | Wire, I = 10 A, L = 2 m (x), B = 0.5 T (y) | 10 N along +z | $10^{-9}$ |
| 2 | Closed loop in a uniform field | Net force 0 | $10^{-9}$ of I·L·B |
| 3 | Long wire, 1 kA at 1 m | $2.000 \times 10^{-4}$ T | $10^{-6}$ |
| 4 | Kepler orbit, a = 1 AU, e = 0.3, 10 periods | Back to start | $10^{-6}$ a; energy $10^{-9}$ |
| 5 | Solar system, 5 years, no push | Energy constant | $10^{-9}$ |
| 6 | Earth–Moon barycentre period | 365.256 days | 0.1 day |
| 7 | Hohmann 1 → 1.524 AU | 2.94 + 2.65 km/s, ~259 days | 0.02 km/s; 2 days |
| 8 | Push $10^{-8}$ m/s² along velocity, 1 year | a grows ~3,170 km | 1% |
| 9 | Cantilever under self-weight | $qL^2/2$ and $qL^4/8EI$ | 1% |
| 10 | Equilibrium temperature, 1 AU, albedo 0.306 | ~254 K | 2 K |

### Example: benchmark 1 in Python

```
import numpy as np
I, L, B = 10.0, np.array([2.0, 0, 0]), np.array([0, 0.5, 0])
F = I * np.cross(L, B)
assert np.allclose(F, [0, 0, 10.0], rtol=1e-9), F
```

### Example: the ledger check

```
# after a propulsion run
dP_bodies = sum(m_i * dv_i for each body)            # change in momentum of all bodies
J_applied = integral of every external force over time
J_partners = momentum given to every reaction partner (plasma, ejecta, light)
assert norm(dP_bodies - J_applied) < 1e-9 * norm(J_applied)   # integrator is honest
assert norm(J_applied + J_partners) < 1e-9 * norm(J_applied)  # nothing created from nothing
```

# 7. Hypothetical enablers and the energy bonus

A hypothetical enabler is one declared assumption that you may treat as true inside your simulation, though today's physics or engineering cannot yet deliver it. You may declare up to three, in writing, at the mid-point check-in. They cannot change afterwards.

**Allowed when bounded and labelled**

- A power source up to a stated wattage, waste heat still radiated
- Ideal energy storage at a stated energy density
- Ideal nanotube or graphene arms (50–130 GPa)
- A room-temperature superconductor up to a stated current density
- The ring and arms already built at the start
- Free raw material delivered where needed
- Ideal motors, sensors and control
- A declared external field carried by a stated plasma
- Unlimited plasma contact up to a stated current
- Access to an asteroid of stated mass

**Never allowed**

- Reactionless thrust: "a force appears on Earth"
- Breaking a conservation law: negative mass, free energy, faster than light
- Switching off gravity, inertia or Earth's magnetic field
- Assuming the outcome: "the climate stays constant"
- Ignoring a load: "the arms carry any stress"

Fewer and smaller enablers score higher under Scientific honesty.

## The energy bonus: up to 10 extra marks

Declare no hypothetical energy source and power your concept from a real, modelled source. Full marks go to a source that supplies all the energy within the time limit; a partial source earns its share.

| Route | Scale for $1.7 \times 10^{33}$ J over 10,000 years | What to model |
|---|---|---|
| Sunlight | About $4 \times 10^{18}$ m² of collectors at 1 AU, some 31,000 times Earth's cross-section | Array mass, build-up, efficiency, shading of Earth, waste heat |
| Fusion (deuterium–tritium) | About $5 \times 10^{18}$ kg of fuel; Earth's oceans hold only about 2% of the deuterium needed | Fuel supply, reactor mass, power conversion, waste heat |
| Your own idea | Show the scale with a calculation first | Every joule traced to its source |

Scales are rounded planning estimates from standard values; check them in your own model.

## 8. Common mistakes that cost marks

| Mistake | Why it is wrong | What to do |
|---|---|---|
| Counting a force on the arms as thrust on Earth | If the reaction lands on Earth's core or structure, it is internal | Book every force with its reaction partner |
| Open circuits | Current must return; the missing path carries the opposite force | Model the full loop |
| Pushing on "the magnetic field" | A field has a source; that source takes the reaction | Say what carries the field and model it |
| 32-bit physics | Earth's position resolves only to ~10 km in single precision | Double precision; track deviations |
| One time step for everything | Vibration needs milliseconds; orbits need hours | Separate clocks for fast and slow physics |
| Ignoring the Moon | A push on Earth alone can leave the Moon behind | Simulate the Moon and check it stays bound |
| Unlabelled visual scaling | Judges cannot tell real size from enlarged | Show the scale factor on screen |
| Hard-coded or tuned results | Leads to disqualification | Every number comes from the running model |
| No convergence check | A result that changes with the step size is not a result | Compare h and h/2 |

# 9. Presenting and submitting

## Submission checklist

| Deliverable | Details |
|---|---|
| Source code | Git repository with a README: setup, run, test |
| Runnable simulation | Executable, web link, or a laptop ready to demo |
| Technical report | At most 8 pages: concept, enablers, models and labels, equations, benchmarks, results, limits |
| Conservation and momentum record | Baseline drift and the full momentum ledger of the propulsion run |
| Demo video | At most 3 minutes |
| Pitch deck | 7-minute pitch, then 5 minutes of questions |

## How you are judged

| Criterion | Marks |
|---|---|
| Physical correctness | 30 |
| Validation and numerical rigour | 20 |
| Scientific honesty | 15 |
| Depth of the simulation | 15 |
| Visualisation and usability | 10 |
| Presentation and questions | 10 |
| Energy bonus (optional) | +10 |

> **A strong 7-minute pitch**
>
> 1. The concept in one picture (30 s).
> 2. Your section-3 estimate and the enablers you chose (1 min).
> 3. Live run: deployment, forces, the ledger (2 min).
> 4. Does Earth reach the orbit, and is it safe? (2 min).
> 5. Benchmarks and what you would test next (1.5 min).

Careers offers are based on individual contribution, so keep a short log of who built what. Every member should be able to explain the code and the physics.

## 10. Resources

| Resource | Use it for |
|---|---|
| JPL Horizons, ssd.jpl.nasa.gov/horizons | Accurate positions and velocities of the Sun, planets and Moon |
| JPL approximate planetary elements, ssd.jpl.nasa.gov/planets/approx_pos.html | Quick initial conditions from orbital elements |
| NASA planetary fact sheets, nssdc.gsfc.nasa.gov/planetary/factsheet | Masses, radii, rotation periods |
| IGRF geomagnetic model, ncei.noaa.gov | Earth's magnetic field coefficients |
| REBOUND, rebound.readthedocs.io | A tested N-body integrator to compare against |
| SciPy solve_ivp, docs.scipy.org | Adaptive integrators in Python |
| Curtis, *Orbital Mechanics for Engineering Students* | Orbits, transfers, perturbations |
| Griffiths, *Introduction to Electrodynamics* | Lorentz force, Biot–Savart, field momentum |
| Timoshenko and Gere, *Mechanics of Materials* | Beams, buckling, stress |

### Questions during the event

Ask the judges at the opening briefing or contact the organisers at [organiser email and phone]. The rulebook is the final word on every rule.

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